

Test Question and Answers solving

In [1]: *# Q. 1. What will be the output of the following code? **

```
x = [1,2,3,4,5]
result = [i**2 for i in x if 1 % 2 != 0]
print(result)
```

[1, 4, 9, 16, 25]

In [2]: *# 2.What will be the output of the following code? **

```
data = [45,72,86,55,91,63,78]
total = 0
count = 0
for num in data:
    if num >= 70:
        total += num
        count += 1
print(round(total/count,2),count)
```

81.75 4

In [9]: *# 3.What will be the output of the following code? **

```
import random
import string

def generate_password(length):
    chars = string.ascii_letters + string.digits + "!@# $"
    password = ''.join(random.choice(chars) for _ in range(length))
    return password

pwd = generate_password(8)
print(len(pwd), type(pwd))
```

8 <class 'str'>

In [10]: *# 4. What will be the output of the following code? **

```
fruits = ['apple', 'bannana', 'cherry', 'mango']
print(fruits[1:3])
```

['bannana', 'cherry']

In []: *# 5. Which of the following statements about Python Lists is CORRECT?*

C.Lists maintain insertion order and allow duplicate elements

In [11]: *# 6.What will be the output of the following code?*

```
numbers = [3,1,4,1,5,9,2,6]
a = sorted(numbers)
print(a[0], numbers[0])

# sorted() creates a new sorted list. a = [1,1,2,3,4,5,6,9] (Its applies sort condition and so on)
# number list is as it numbers = [3,1,4,1,5,9,2,6]
```

1 3

In [15]: *# 7.What will be the output of the following code?*

```
student = {'name': 'Alice', 'age': 22, 'grade': 'A'}
```

```

student['age'] = 23
student['city'] = 'Hyderabad'
print(len(student), student.get('grade'), student.get('score', 0))

```

4 A 0

In [16]: # 8.What will be the output of the following code?

```

marks = {'Maths': 85, 'Science': 90, 'English': 78}
marks['Science'] = 95
del marks['English']
for key, value in marks.items():
    print(key, value, end= ' | ')

```

Maths 85 | Science 95 |

In [19]: # 9.What will be the output of the following code?

```

t = [10,20,30,40,50]
a,b,*c = t
print(a,b,c)

```

10 20 [30, 40, 50]

In [21]: # 10.What will be the output of the following code?

```

t1 = (1,2,3)
t2 = (3,4,5)
t3 = t1 + t2
print(t3.count(3), t3.index(4), len(t3))

```

count(3) → How many 3's? → 2

index(4) → Where is 4? → 4

len(t3) → Total elements? → 6

2 4 6

In [22]: # 11.What will be the output of the following code?

```

s = {1,2,3,2,1,4}
s.add(5)
s.discard(2)
s.discard(10)
print(len(s), 3 in s)

```

A set stores only unique values. Duplicates are automatically removed. s = {1,2,3,4}

we add 5 to the set .s = {1,2,3,4,5}

discard 2,10 there is no value 10 so set is, s = {1,3,4,5} now apply print condition

(len(s), 3 in s) len s is 4 and it finds existence of 3 value in set i.e true (3 is available)

4 True

In [23]: # 12.What will be the output of the following code?

```

a = {1, 2,3,4,5}
b = {3,4,5,6,7}
print(a & b, a - b) # The & (intersection) operator finds elements present in both sets. It returns a set.
# Set stores only unique values. Duplicates are automatically removed. a & b = {3,4,5}
# while a - b returns elements present in the first set but not in the second set (difference, it returns a set)

```

{3, 4, 5} {1, 2}

In [24]: # 13.What will be the output of the following code?

```
s = 'Hello Data Science'
print(s.split()[1], s.lower().count('e'), s.replace('Science', 'World'))

# Here Split()[1] is Data (as per index) count 'e' = 3 Science is replaced with World
```

Data 3 Hello Data World

In [26]: # 14.What will be the output of the following code?

```
s = 'Python Programming'
print(s[7:18:2], s[::-1])
```

Pormig gnimmargorP nohtyP

In [28]: # 15.What will be the output of the following code?

```
words = ['Python', 'is', 'fun']
sentence = '-'.join(words)
parts = sentence.split('-')
print(sentence, len(parts))
```

Python-is-fun 3

In [29]: # 16.What will be the output of the following code?

```
x = 5
y = 10
z = 15
print(x < y and y < z or z < x )
```

True

In [30]: # 17.What will be the output of the following code?

```
num = 28
if num % 2 == 0:
    if num % 7 == 0:
        print('Even and Divisible by 7')
    else:
        print('Even but not divisible by 7')
else:
    print('Odd number')
```

Even and Divisible by 7

In [31]: # 18.What will be the output of the following code?

```
num = 29
is_prime = True

if num < 2:
    is_prime = False
elif num == 2:
    is_prime = True
elif num % 2:
    is_prime == 0

else:
    if num % 3 == 0 or num % 5 == 0 or num % 7 == 0:
        is_prime = False

if is_prime:
    print(f'{num} is prime')
```

```
else:
    print(f'{num} is not prime')
```

29 is prime

In [33]: *# 19.What will be the output of the following code?*

```
a,b,c = 10,20,30
if a > b or b < c:
    if a + b > c:
        print('X')
    elif a * 2 == b:
        print('Y')
    else:
        print('Z')
else:
    print('W')
```

Y

In [34]: *# 20.What will be the output of the following code?*

```
char = 'E'

if char.isalpha():
    if char.lower() in 'aeiou':
        if char.isupper():
            print('Uppercase Vowel')
        else:
            print('Lowercase Vowel')
    else:
        if char.isupper():
            print('Upper Consonent')
        else:
            print('Lower Consonent')
else:
    print('Not an Alphabet')
```

Uppercase Vowel

In [40]: *# 21.What will be the output of the following code?*

```
for i in range (1,6):
    if i == 3:
        pass
    print(i, end = ' ')
```

1 2 3 4 5

In [41]: *# 22.What will be the output of the following code?*

```
count = 0

for i in range (1,21):
    if i % 2 == 0:
        continue
    if i % 7 == 0:
        break
    count += 1
print(count)

# continue skips even numbers.
# break stops the loop when a number divisible by 7 is found.
# count increases only for odd numbers before reaching 7.
# Final output is 3.
```

In [45]: # 23.What will be the output of the following code?

```
vowels = 0
consonants = 0
word = 'Programming'

for char in word:
    if not char.isalpha():
        pass
    elif char.lower() in 'aieou':
        vowels += 1
    else:
        consonants += 1
print(f'Vowels: {vowels}, Consonants: {consonants}')
```

Vowels: 3, Consonants: 8

In [46]: # 24.What will be the output of the following code?

```
for num in range (1,6):
    match num % 2:
        case 0:
            print(f'{num} Even', end= '|')
        case 1:
            print(f'{num} Odd', end= '|')
        case _:
            print('Unknown', end= '|')
```

1 Odd|2 Even|3 Odd|4 Even|5 Odd|

In []: # 25 .Which of the following correctly describes the difference between np.arange() and np.linspace()

A) np.arange() generates values based on step size while np.linspace() generates fixed number

In [47]: # 26.What will be the output of the following code?

```
import numpy as np

a = np.array([10,20,30,40,50])
b = np.zeros((2,3), dtype = int)
c = np.ones((2,3), dtype = int)
print(a[1:4].sum(), b.shape, (b + c).sum())
```

90 (2, 3) 6

In [51]: # 27.What will be the output of the following code?

```
import numpy as np

a = np.linspace(0,100,5) # linspace: It creates numbers that are equally spaced between a start and end value
print(a) # Start = 0, End = 100, Total values = 5 # (100 - 0) / (5 - 1) = 100 / 4 = 25
```

[0. 25. 50. 75. 100.]

In [57]: # 28. What will be the output of the following code?

```
import pandas as pd

data = {'Name': ['Alice', 'Bob', 'Charlie', 'David'],
        'Age': [25,30,35, 28],
        'Score': [85,90,78, 92]}
df = pd.DataFrame(data)
print(df[df['Score'] > 85] ['Name'].values)
```

['Bob' 'David']

In [59]: # 29.What will be the output of the following code?

```
import pandas as pd

df = pd.DataFrame({'City': ['Hyd', 'Delhi', 'Hyd', 'Mumbai', 'Delhi'], # Sort will be applied
                  'Sales':[ 100, 200, 150, 300, 250]})

result = df.groupby('City')['Sales'].sum().reset_index()
print(result.sort_values('Sales', ascending = False)['City'].values[0])
```

Delhi

In [60]: # 30.What will be the output of the following code?

```
import pandas as pd

df = pd.DataFrame({'Color': ['Red','Blue', 'Red', 'Green'],
                  'Price': [100,200,150,300]})

df_encoded = pd.get_dummies(df, columns = ['Color'])
print(df_encoded.shape, df_encoded.columns.tolist())
```

(4, 4) ['Price', 'Color_Blue', 'Color_Green', 'Color_Red']

In [62]: # 31.What will be the output of the following code?

```
import pandas as pd
import numpy as np

data = {'Name': ['Alice', 'Bob', 'Charlie', 'David'],
        'Score': [85,np.nan,78, 92]}
df = pd.DataFrame(data)
df['Score'].fillna(df['Score'].mean(), inplace = True)
print(round(df['Score'].sum(), 1))
```

340.0

C:\Users\user\AppData\Local\Temp\ipykernel_24656\558627754.py:9: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.

The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.

```
df['Score'].fillna(df['Score'].mean(), inplace = True)
```

In []: # 32.Which of the following is the CORRECT first step in Exploratory Data Analysis?

B) Understand the dataset – check shape, data types, missing values and basic statistics

In []: # 33.What is the primary purpose of a heatmap in Seaborn?

B. To visualize relationships or correlations between variables using colors

In []: # 34.Why are visualizations created using Matplotlib important during data analysis?

B. They help identify patterns, trends, distributions, and outliers in the data.

In []: # 35.What is the primary purpose of a box plot in data analysis?

B. To visualize the distribution of data, identify the median, quartiles, and detect potential outliers

In []:

In []: